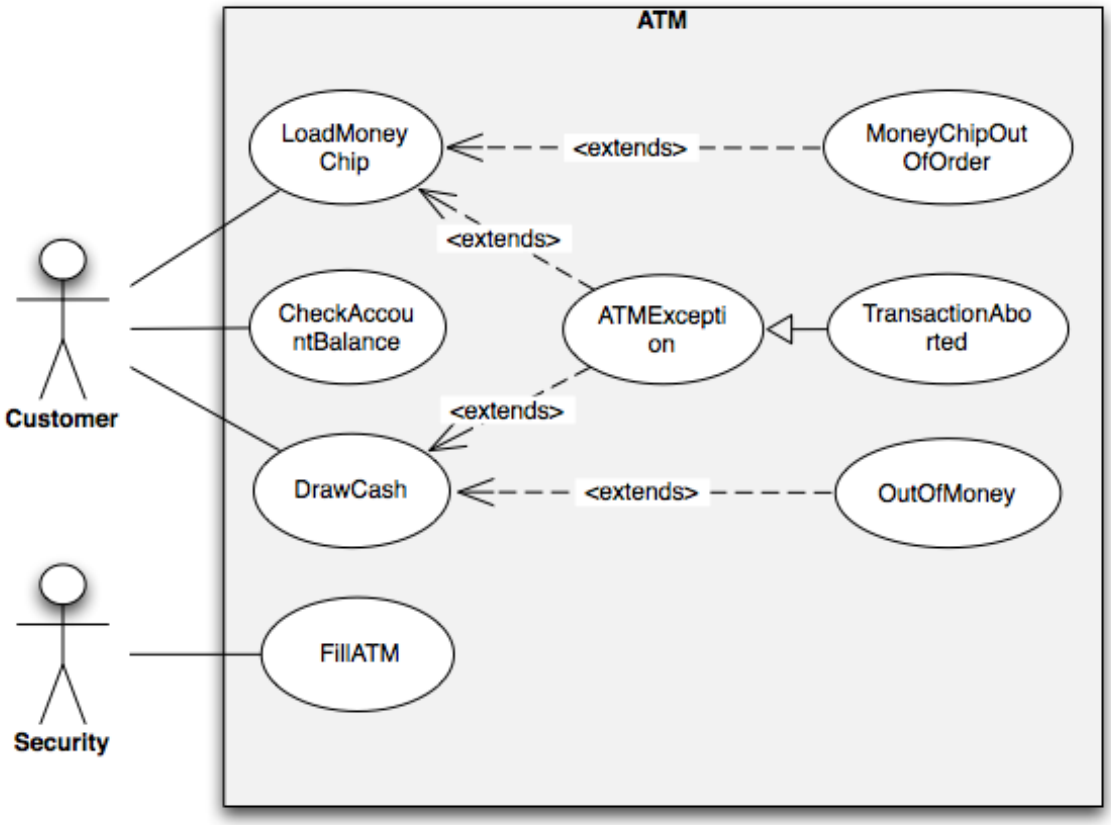


Object-Oriented Software Engineering:
Using UML, Patterns and Java, 3rd Ed.

Q1. Model an ATM

Draw a use case diagram for an ATM (cash terminal). The system includes two actors: a customer, who draws money from his account either on the money chip on his bank card or in cash, and a security man, who fills money into the ATM.
Use cases should include: DrawCash, LoadMoneyChip, CheckAccountBallance, FillATM. Also include the following exceptional cases: OutOfMoney, TransactionAborted (i.e., customer selected the cancel button without completing the transaction) and MoneyChipOutOfOrder.
Remark: It is possible to use inheritance between use cases!

Write the flow of events and specify all fields for the use case CheckAccountBalance that you drew before. Specify relationships.



Use case name	CheckAccountBalance		
Participating Actors	Initiated by Customer		
Flow of events	1. The Customer inserts his bank card into the ATM.		
		2. The ATM welcomes the Customer and opens the main menu.	
		3. The ATM waits for the selection from the Customer.	
	4. The Customer selects the “CheckAccountBalance” function by pressing the appropriate button.		
		5. The ATM system shows the enter PIN screen to the Customer.	
	7. The Customer enters his correct PIN and confirms it.		
		5. The ATM system shows the actual balance of the Customer’s account.	
Entry condition	The ATM is working and Customer is in front of the terminal.		
Exit condition	The Customer can view his account balance.		
Quality requirements	The ATM stays online for the whole period of the CheckAccountBalance use case.		

Q2. Model a book

Draw a class diagram representing a book defined by the following statement: “A book is composed of a number of parts, which in turn are composed of a number of chapters. Chapters are composed of sections.” First, focus only on classes and associations.

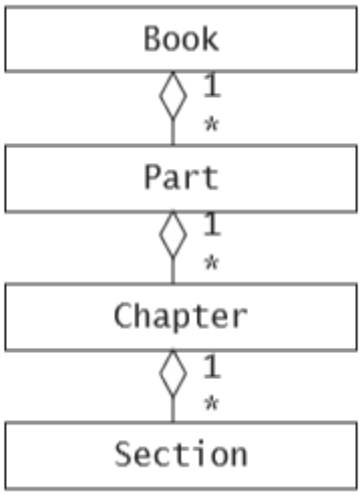
Add multiplicity to the class diagram you produced.

Refine the class diagram to include the following attributes:

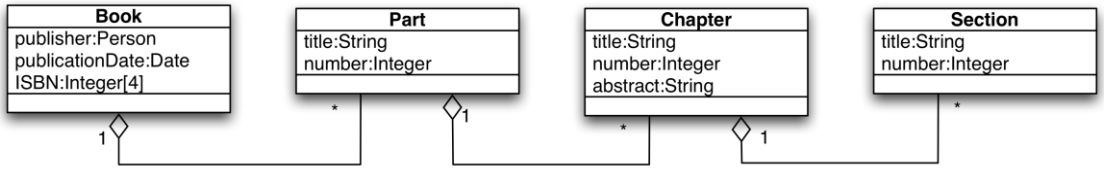
- Book includes a publisher, publication date, and an ISBN
- Part includes a title and a number
- Chapter includes a title, a number, and an abstract
- Section includes a title and a number

Consider the refined class diagram. Note that the Part, Chapter, and Section classes all include a title and a number attribute. Use inheritance to factor out these two attributes .

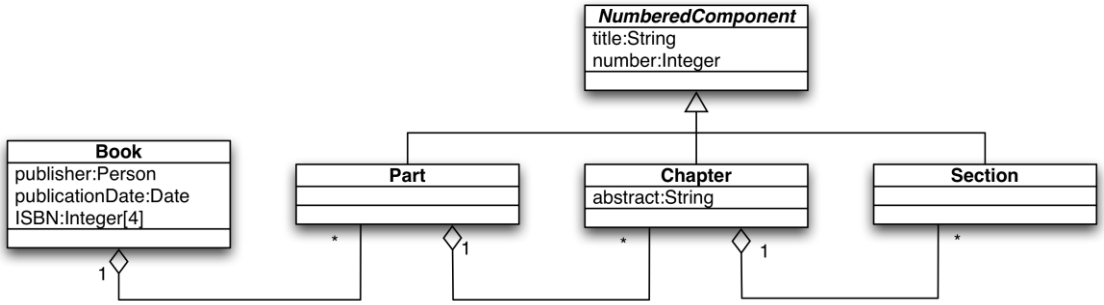
1.)



2.)



3.)



Q3. Happy Hour Message

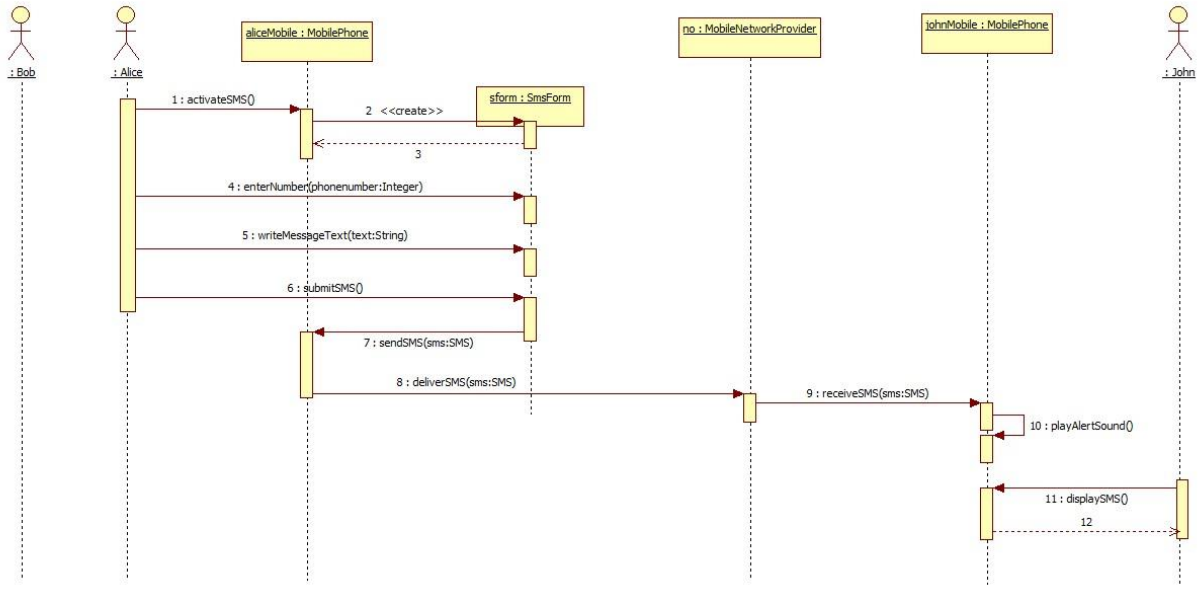
Draw a sequence diagram for the HappyHourMessage scenario in the figure 1 below. Draw only the first five messages exchanged.

Scenario name	HappyHourMessage
Participating actor instances	bob, alice: BarFlies john: Friend
Flow of events	1. Bob and Alice are sitting in their favorite pub and it's happy hour. They want to invite their common friend John who likes cocktails and didn't know about the newly scheduled happy hour. Alice takes out her "mobile phone" and activates the "SMS" function.

	2. Alice enters John's cell phone number and writes the message about the happy hour into the "SMS text field". She sends the message and waits for John's answer. 3. John, who is still at work, is alerted by a sound of his cell phone that a new "SMS" has arrived. He reads the lines from Alice and answers that he will come immediately. He quits working and leaves the office. 4. Alice receives John's answer at her "mobile phone".

Figure 1

Solution:



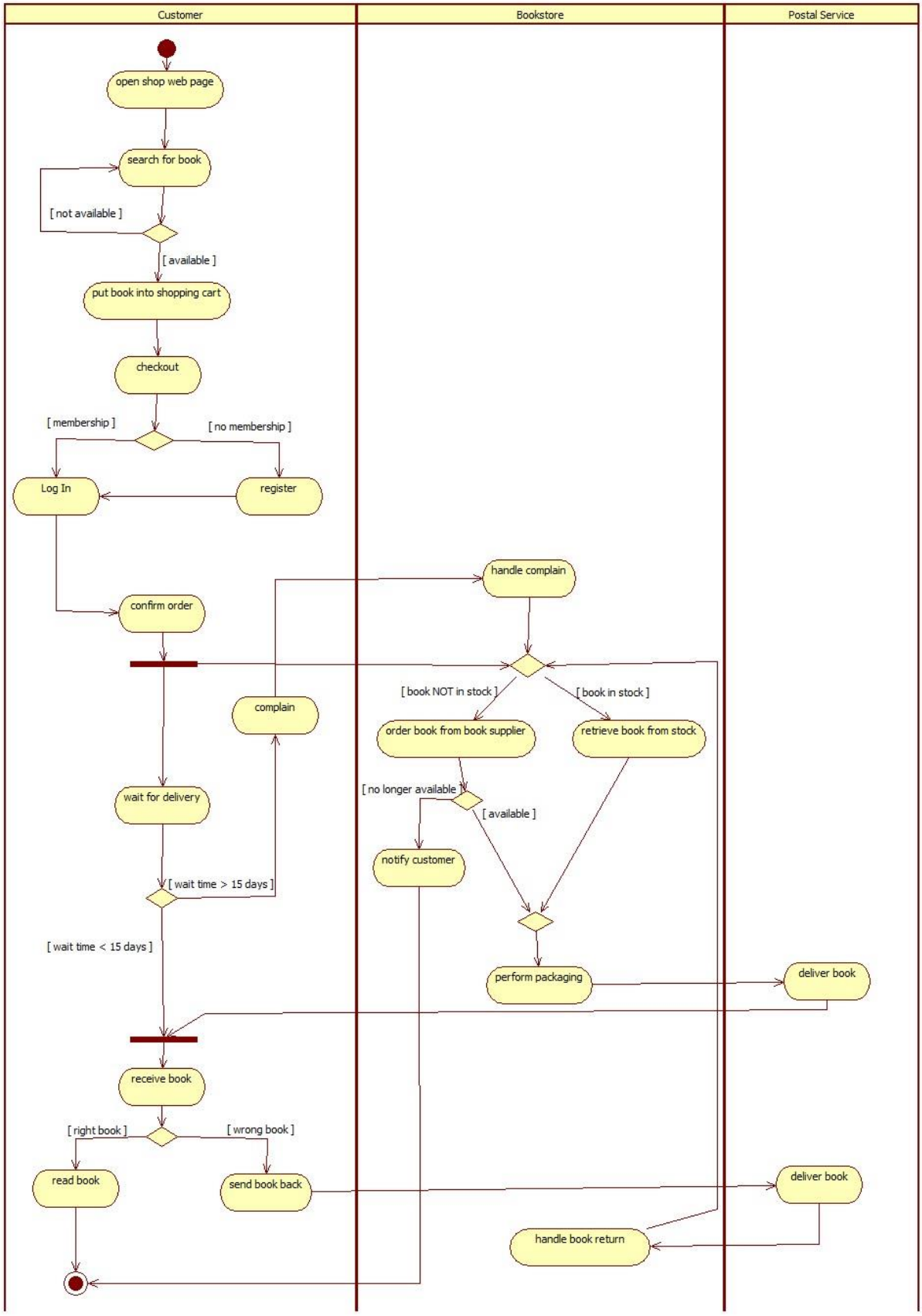
Q4. Model an online bookstore

Consider the process of ordering a book via Internet. Draw an activity diagram representing each step of the process, from the moment you open the shop's web page to the point when you start reading the book, including login or registration. Include activities that others need to perform.

Add additional exception handling to the activity diagram you have developed. Model the following three exceptions:

1. The order got lost in the Internet
2. The wrong book is packaged
3. The ordered book is no longer available

Solution:


Customer
Bookstore
Postal Service
open shop web page
search for book
[not available]
[available]
put book into shopping cart
checkout
[membership]
[no membership]
Log In
register
confirm order
handle complain
[book NOT in stock]
[book in stock]
complain
order book from book supplier
retrieve book from stock
wait for delivery
[no longer available]
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perform packaging
deliver book
receive book
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read book
send book back
deliver book
handle book return